



Green University of Bangladesh
Department of Computer Science and Engineering
(CSE)

Faculty of Sciences and Engineering
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LAB REPORT NO: 3

Course Title: Web Programing Lab

Course Code: CSE 302

Section: 232-D2

Lab Experiment Name: Interactive web page to load new data on-the-fly from a JSON File in a tabular form Using AJAX and some CSS.

Student Details

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Lab Report Status

Marks:

Comments:.....

Signature:.....

Date:.....

1. Introduction:

This experiment demonstrates how asynchronous web technologies can be used to dynamically load external data into a webpage without requiring a page reload. Using AJAX (Asynchronous JavaScript and XML) and a publicly available JSON dataset of U.S. Presidents, the web page retrieves data on-the-fly and displays it in a structured tabular format. The purpose of this task is to show how client-side JavaScript can communicate with a server, fetch JSON data from a remote URL, parse it, and update the Document Object Model (DOM) dynamically. This approach is widely used in modern web applications for creating interactive and responsive user interfaces. By completing this exercise, we gain practical experience with asynchronous requests, JSON handling, and the integration of HTML, CSS, and JavaScript to build data-driven web pages.

2. OBJECTIVES/AIM

The main objectives of this experiment are:

1. To understand how AJAX enables asynchronous communication between a web page and a server.
2. To learn how to retrieve external data on-the-fly using the XMLHttpRequest object.
3. To load and parse data from a JSON file stored at a remote URL.
4. To dynamically update the web page without reloading, using JavaScript DOM manipulation.
5. To display the retrieved JSON data in a well-formatted HTML table.
6. To apply basic CSS styling to enhance the visual presentation of dynamically loaded data.
7. To gain practical experience in integrating HTML, CSS, and JavaScript to build interactive and responsive web pages.

3. PROCEDURE

1. Setting up the Project
 - index.html for the structure of the page
 - style.css for styling the page
 - ajax-presidents.js for JavaScript functionality

2. HTML Page Creation

A basic structure was built with:

- Style the page layout, table, and button.
- Add hover effects, borders, and alternating row colors to the table for improved readability and aesthetics.

3. CSS Styling

The style.css file was used to:

- A header displaying the title "US Presidents (AJAX + JSON)"
- A button labeled "Load Presidents Data" to trigger the data fetch
- An empty HTML table with headers for displaying the JSON data

4. Implementing AJAX with JavaScript

- A new instance of XMLHttpRequest was created in the ajax-presidents.js file.
- The open() method was called with the URL for the JSON file, followed by send() to initiate the request.
- The request was asynchronous, and a callback function was set up to handle the response

5. Handling JSON Data

- Once the response was received, the data was parsed from JSON format using JSON.parse().
- The parsed data, which contained an array of president details, was looped through using a for loop.

6. Dynamically Inserting Data into the Table

- A JavaScript function dynamically created table rows from the parsed JSON data.
- These rows were inserted into the table body (<tbody>) of the HTML using the innerHTML method.

7. Testing the Functionality

- The webpage was loaded in a browser, and the "Load Presidents Data" button was clicked to fetch the JSON data.
- Upon clicking the button, the data loaded instantly, without page refresh, and populated the table with the required information.

4.Implementation:

Source Code:

index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta http-equiv="X-UA-Compatible" content="IE=edge">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>AJAX lab report 3 Topu</title>
```



```
<script src="ajax-presidents.js"></script>

</body>

</html>
```

style.css

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: Arial, Helvetica, sans-serif;
}

body {
  padding: 20px;
  background-color: #f4f4f4;
}

header {
  margin-bottom: 15px;
}

h1 {
  font-size: 26px;
  margin-bottom: 10px;
  color: #333;
}

#loadBtn {
  padding: 10px 18px;
  border: none;
  background-color: rgb(70, 138, 207);
  color: #c2ce16;
  font-size: 15px;
  cursor: pointer;
  border-radius: 4px;
  transition: background-color 0.2s ease;
}

#loadBtn:hover {
```

```
        background-color: #006666;
    }

    #loadBtn.btn-disabled {
        background-color: #888;
        cursor: not-allowed;
    }

    .table-wrapper {
        margin-top: 20px;
        overflow-x: auto;
    }

    /* Table styling */
    table {
        width: 100%;
        border-collapse: collapse;
        background-color: #fff;
    }

    th,
    td {
        border: 1px solid #ddd;
        padding: 8px 10px;
        text-align: center;
        font-size: 14px;
    }

    th {
        background-color: #f0f0f0;
        font-weight: bold;
    }

    tr:nth-child(even) {
        background-color: #fafafa;
    }

    tr:hover {
        background-color: #cb1729;
    }
}
```

[ajax-presidents.js](#)

```
var loadBtn = document.getElementById("loadBtn");
var tableBody = document.getElementById("presidentsBody");
var dataLoaded = false;

loadBtn.addEventListener("click", function () {
    if (dataLoaded) {
        return;
    }

    var xhr = new XMLHttpRequest();

    xhr.open(
        "GET",
        "https://raw.githubusercontent.com/hitch17/sample-data/master/presidents.json",
        true
    );

    xhr.onreadystatechange = function () {
        if (xhr.readyState === 4) {
            if (xhr.status === 200) {
                var presidents = JSON.parse(xhr.responseText);
                renderTable(presidents);
                dataLoaded = true;

                loadBtn.textContent = "Data Loaded";
                loadBtn.disabled = true;
                loadBtn.classList.add("btn-disabled");
            } else {
                alert("Error loading data. Status: " + xhr.status);
            }
        }
    };

    xhr.send();
});

// Create table rows from JSON data
```

```
function renderTable(data) {
    var rowsHtml = "";

    for (var i = 0; i < data.length; i++) {
        var p = data[i];

        rowsHtml +=
            "<tr>" +
            "<td>" + (p.number || "") + "</td>" +
            "<td>" + (p.president || "") + "</td>" +
            "<td>" + (p.birth_year || "") + "</td>" +
            "<td>" + (p.death_year === null ? "" : p.death_year) +
            "</td>" +
            "<td>" + (p.took_office || "") + "</td>" +
            "<td>" + (p.left_office || "") + "</td>" +
            "<td>" + (p.party || "") + "</td>" +
            "</tr>";
    }

    tableBody.innerHTML = rowsHtml;
}
```

Outputs:

Empty table body

US Presidents (AJAX + JSON)

Load Presidents Data

No	President	Birth Year	Death Year	Took Office	Left Office	Party
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Load Data:

US Presidents (AJAX + JSON)

Data Loaded

No	President	Birth Year	Death Year	Took Office	Left Office	Party
1	George Washington	1732	1799	1789-04-30	1797-03-04	No Party
2	John Adams	1735	1826	1797-03-04	1801-03-04	Federalist
3	Thomas Jefferson	1743	1826	1801-03-04	1809-03-04	Democratic-Republican
4	James Madison	1751	1836	1809-03-04	1817-03-04	Democratic-Republican
5	James Monroe	1758	1831	1817-03-04	1825-03-04	Democratic-Republican
6	John Quincy Adams	1767	1848	1825-03-04	1829-03-04	Democratic-Republican
7	Andrew Jackson	1767	1845	1829-03-04	1837-03-04	Democratic
8	Martin Van Buren	1782	1862	1837-03-04	1841-03-04	Democratic
9	William Henry Harrison	1773	1841	1841-03-04	1841-04-04	Whig

10	John Tyler	1790	1862	1841-04-04	1845-03-04	Whig
11	James K. Polk	1795	1849	1845-03-04	1849-03-04	Democratic
12	Zachary Taylor	1784	1850	1849-03-04	1850-07-09	Whig
13	Millard Fillmore	1800	1874	1850-07-09	1853-03-04	Whig
14	Franklin Pierce	1804	1869	1853-03-04	1857-03-04	Democratic
15	James Buchanan	1791	1868	1857-03-04	1861-03-04	Democratic
16	Abraham Lincoln	1809	1865	1861-03-04	1865-04-15	Republican
17	Andrew Johnson	1808	1875	1865-04-15	1869-03-04	Democratic
18	Ulysses S. Grant	1822	1885	1869-03-04	1877-03-04	Republican
19	Rutherford B. Hayes	1822	1893	1877-03-04	1881-03-04	Republican
20	James A. Garfield	1831	1881	1881-03-04	1881-09-19	Republican
21	Chester A. Arthur	1829	1886	1881-09-19	1885-03-04	Republican
22	Grover Cleveland	1837	1908	1885-03-04	1889-03-04	Democratic
23	Benjamin Harrison	1833	1901	1889-03-04	1893-03-04	Republican

25	William McKinley	1843	1901	1897-03-04	1901-09-14	Republican
26	Theodore Roosevelt	1858	1919	1901-09-14	1909-03-04	Republican
27	William Howard Taft	1857	1930	1909-03-04	1913-03-04	Republican
28	Woodrow Wilson	1856	1924	1913-03-04	1921-03-04	Democratic
29	Warren G. Harding	1865	1923	1921-03-04	1923-08-02	Republican
30	Calvin Coolidge	1872	1933	1923-08-02	1929-03-04	Republican
31	Herbert Hoover	1874	1964	1929-03-04	1933-03-04	Republican
32	Franklin D. Roosevelt	1882	1945	1933-03-04	1945-04-12	Democratic
33	Harry S. Truman	1884	1972	1945-04-12	1953-01-20	Democratic
34	Dwight D. Eisenhower	1890	1969	1953-01-20	1961-01-20	Republican
35	John F. Kennedy	1917	1963	1961-01-20	1963-11-22	Democratic
36	Lyndon B. Johnson	1908	1973	1963-11-22	1969-01-20	Democratic
37	Richard Nixon	1913	1994	1969-01-20	1974-08-09	Republican
38	Gerald Ford	1913	2006	1974-08-09	1977-01-20	Republican

34	Dwight D. Eisenhower	1890	1969	1953-01-20	1961-01-20	Republican
35	John F. Kennedy	1917	1963	1961-01-20	1963-11-22	Democratic
36	Lyndon B. Johnson	1908	1973	1963-11-22	1969-01-20	Democratic
37	Richard Nixon	1913	1994	1969-01-20	1974-08-09	Republican
38	Gerald Ford	1913	2006	1974-08-09	1977-01-20	Republican
39	Jimmy Carter	1924		1977-01-20	1981-01-20	Democratic
40	Ronald Reagan	1911	2004	1981-01-20	1989-01-20	Republican
41	George H. W. Bush	1924	2018	1989-01-20	1993-01-20	Republican
42	Bill Clinton	1946		1993-01-20	2001-01-20	Democratic
43	George W. Bush	1946		2001-01-20	2009-01-20	Republican
44	Barack Obama	1961		2009-01-20	2017-01-20	Democratic
45	Donald J. Trump	1946		2017-01-20	2021-01-20	Republican
46	Joseph 'Joe' Biden	1942		2021-01-20		Democratic

6. ANALYSIS AND DISCUSSION:

This experiment demonstrates the use of AJAX (Asynchronous JavaScript and XML) to load JSON data dynamically from an external source without reloading the page. Using XMLHttpRequest, an asynchronous request is sent to the server to fetch the data. Once the data is retrieved, JSON.parse() is used to convert the raw JSON text into a JavaScript object.

The parsed data is then dynamically inserted into an HTML table using JavaScript. Each president's information, including name, birth year, and political party, is displayed in table rows that are generated in real-time. This approach ensures that the page remains responsive and interactive without the need for a full reload.

CSS is applied to improve the page's design, making the table more readable and visually appealing. Alternating row colors and hover effects are used to enhance the user interface.

While the experiment works as expected, future improvements could include error handling for network failures and pagination for handling larger datasets. These enhancements would make the application more robust and scalable for real-world use.

Overall, the experiment successfully demonstrates how AJAX and JSON can be used to create dynamic, interactive web pages with real-time content updates.